



Vandan Nagori

Roll No.:230021020

B.Tech

Space Science and Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. SSE	Indian Institute of Technology Indore	8.58 (Current)	2023-Present
Senior Secondary	CBSE Board	93.2%	2022
Secondary	CBSE Board	86.2%	2020

EXPERIENCE

- PRIUS Fellowship** *Office Memorandum*
 - Worked under PRIUS program to classify galaxies as passive or star-forming using COSMOS survey data.
 - Used **EAZY-py** for photometric redshift estimation via template fitting and generated UVJ diagrams.
 - Performed preprocessing, parameter tuning, and output analysis for redshift and galaxy-type classification.
- ISRO-NRSC Challenge** *Output, Certificate*
 - Participated in ISRO-NRSC national challenge for cloud and shadow detection using TOA reflectance data.
 - Performed preprocessing (reflectance conversion, masking) and applied UNet-based ML segmentation.
 - Developed and optimized training pipelines for accurate detection and evaluation.

PROJECTS

- Immersive 3D Portfolio (Space-Themed)** *Website*
 - Architected an interactive 3D portfolio using React Three Fiber and Three.js featuring animated spacecraft models with launch sequences, holographic UI panels, and explorable solar system navigation for immersive user experience.
 - Optimized WebGL rendering performance through Draco/GLTF compression, progressive asset loading, and code-splitting; reduced bundle size by 40% and maintained consistent 60 FPS on mid-range devices through efficient scene management.
 - Developed custom GLSL shaders for visual effects (engine glow, particle systems), integrated HDR environment lighting, and implemented procedural starfield generation while ensuring cross-platform compatibility.
 - Built reusable component library with smooth scene transitions, gesture-safe touch controls, and complex animation sequences including spacecraft docking mechanics and camera interpolation systems.
- Full-Stack Rental Platform (Airbnb Clone)** *Website, GitHub*
 - Developed a rental platform with Node.js, Express.js, MongoDB, and EJS featuring user authentication, property management, and review system with complete CRUD operations.
 - Architected RESTful APIs following MVC pattern with comprehensive validation layer (server and client-side); integrated Cloudinary for dynamic image upload, storage, and manipulation of property listings.
 - Implemented secure authentication/authorization with passport.js, session management, and role-based access control for property owners and users.
 - Deployed on Render with MongoDB Atlas cloud database; configured CI/CD pipeline and utilized Git/GitHub for version control and collaborative development.
- Gaming Club Website (IIT Indore)** *Website, GitHub*
 - Led the development of the Gaming Club website using React.js, Node.js, Express.js, HTML, CSS, and JavaScript.
 - Deployed and maintained the website using Git and Vercel, ensuring consistent uptime and performance.
 - Collaborated with the club team to implement dynamic content updates and user-friendly interface improvements.
- Astronomy Club Website (IIT Indore)** *Website, GitHub*
 - Built and deployed the official Astronomy Club website using React.js, Node.js, and hosted it on Vercel.
 - Developed responsive UI components and ensured cross-device compatibility for smooth user experience.
 - Collaborated with team members to regularly update content and integrate new sections as per event needs.
- Academic Management System** *GitHub*
 - Built a comprehensive full-stack academic portal using Node.js, Express.js, MongoDB, and EJS with role-based dashboards for faculty and students with distinct access controls.
 - Developed student portal featuring timetable management, quiz system, grade tracking, and homework submissions with real-time updates and organized dashboard interface.
 - Created faculty portal enabling content management system for uploading study materials, creating quizzes, assigning homework, and managing student resources with instant synchronization.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, *DSA
- **Web Dev:** HTML, CSS, Javascript, ReactJS, NodeJS, Express, EJS, Bootstrap, Tailwind CSS, Three.js*
- **Databases:** MongoDB, SQL
- **Tools:** Git/GitHub, Terminal * Elementary proficiency

KEY COURSES TAKEN

- **Mathematics:** Linear Algebra, Differential Equations, Complex Analysis, Numerical Methods and Calculus
- **Core:** Digital Electronics, Analog Electronics, Electronics Lab
- **CS:** Scientific Computing (python), Computer Programming

CERTIFICATIONS

- **Web Development (Delta Batch) – Apna College** Certificate
 - Completed structured full-stack development course: HTML, CSS, JavaScript, Bootstrap, Git/GitHub, Node.js, Express.js, EJS, SQL, MongoDB, React.js, Tailwind CSS, and Redux.
 - Developed full-stack project utilizing RESTful APIs, CRUD operations, and modern web technologies.

POSITIONS OF RESPONSIBILITY

- **Head of Web Development,** Astronomy Club, IIT Indore Jul. 2023 - Jul. 2025
 - **Head of Technicals,** Gaming Club, IIT Indore Dec. 2024 - Apr. 2025
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